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Project Title: Technical Analysis and Response for Latency-Critical Networks to Advance Autonomous Driving

1. Motivation

1.1. Industry Trends: Evolution of Ultra-Reliable Low-Latency Communication (URLLC) based on 5G-Advanced and 6G

The rapid advancements in next-generation communication technologies are fundamentally transforming the paradigm of the autonomous driving industry. The networking industry is transitioning beyond simple data transfer efficiency to latency-critical networks (LCNs) that must meet strict time constraints. In particular, Ultra-Reliable Low-Latency Communications (URLLC), established through 3GPP Releases 16 and 17, are the driving force guaranteeing latency of less than 1 ms and reliability above 99.999% required by autonomous vehicles. This trend continues from 5G-Advanced (Release 18) toward 6G, where networks evolve to integrate sensing and communication.

Global automakers and telecommunications equipment companies now define the vehicle as a 'mobile data center'. Hyundai Motor and Hyundai Mobis have commercialized Level 3 or higher autonomous driving technologies by integrating 5G and Vehicle-to-Everything (V2X) solutions, enabling advanced functions such as Highway Driving Pilot (HDP). These systems must process multi-gigabit data generated in real time by numerous sensors inside the vehicle (LiDAR, cameras, radar) and instantly share it with surrounding infrastructure; hence, the adoption of deterministic networking technologies is essential.

1.2. Practical Relevance: Ensuring Autonomous Driving Safety and the Integration of Edge Computing

In the autonomous driving environment, network performance is not just about convenience; it is a critical safety factor directly linked to passenger life. Existing best-effort networks cannot avoid packet delay or loss during congestion, which poses a risk of collapsing the control loop of autonomous vehicles. In real-world scenarios, when driving at speeds above 100 km/h, even a delay of a few milliseconds causes several meters of difference in braking distance, serving as a decisive variable that determines whether an accident occurs.

To solve these problems, Multi-access Edge Computing (MEC) has become an essential component of the autonomous driving infrastructure. MEC processes data at the network edge instead of sending it to central cloud servers, dramatically reducing the round-trip time (RTT). Global IT leaders such as AWS, Google and Cisco are building architectures that combine private 5G with MEC in collaboration with telecommunications operators such as Verizon, proving the practicality of autonomous driving in real industrial fields such as smart ports, automated guided vehicles (AGV) and intelligent transportation systems (ITS).

1.3. Value for Professional Competencies: Acquiring Practical Knowledge for Next-Generation Network Experts

By mastering the design principles and protocol mechanisms of latency-critical networks through this project, we secure top-tier technical competencies demanded in the modern networking and IT industry. Deeply understanding industrial standards such as IEEE 802.1 TSN (Time-Sensitive Networking), IETF DetNet (Deterministic Networking) and 3GPP URLLC, beyond textbook basics, provides distinctive competitiveness in technical interviews and the workplace.

In particular, network engineers in the field of autonomous driving need to present solutions to complex problems such as “How do we overcome the non-deterministic characteristics of wireless channels?” or “How do we maintain precise time synchronization across heterogeneous networks?” Network slicing, software-defined networking (SDN) and AI-based autonomous network operation technologies are among the most sought-after job-ready skills in today’s IT job market. By covering these cutting-edge technologies in an integrated way, this research helps team members cultivate the ability to solve complex networking challenges in real industrial settings.

2. Problem Definition

2.1. Key Challenges: Ensuring Deterministic Performance in Non-Deterministic Environments

The greatest technical barrier facing today’s autonomous driving networks is to offer strict deterministic performance amid the uncertainties of wireless communication environments. Traditional Ethernet and wireless LAN rely on collision avoidance (CSMA/CA) or statistical multiplexing for best-effort transmission, but they have structural limitations that allow latency to grow indefinitely when traffic is congested. Such uncertainty becomes a critical risk factor when an autonomous vehicle receives collision avoidance commands at an intersection.

In addition, as the vehicle’s electrical/electronic (E/E) architecture evolves to a zone-based configuration, safety data, sensor data and infotainment traffic coexist on a single high-speed Ethernet backbone. To ensure that large volumes of low-priority data do not interfere with the transmission of important control commands, sophisticated implementation of traffic shaping and frame preemption technologies is required.

2.2. Scope of Inquiry and Analysis: Integration of Wired and Wireless Networks and Extension to Layer 3

This study sets the following detailed technical domains as the scope of exploration for implementing LCNs for autonomous driving.

Wired TSN mechanisms (Layer 2): We analyze ways to minimize latency and maximize reliability in in-vehicle networks through functions such as IEEE 802.1Qbv (Time-Aware Shaper), 802.1Qbu (Frame Preemption) and 802.1CB (Replication and Elimination). We specifically reflect the latest content of the 2025 IEEE 802.1DG automotive TSN profile.

Integration of 5G URLLC and wireless TSN: We delve into the structure of DS-TT (Device-side TSN Translator) and NW-TT (Network-side TSN Translator), which bridge low-latency technologies defined in 3GPP standards (mini-slots, grant-free uplink) with wired TSN networks.

IETF DetNet (Layer 3): We analyze the architecture that guarantees bounded latency in multi-hop wide-area networks and techniques for improving reliability through the Packet Replication and Elimination Function (PREOF).

Edge intelligence and autonomous management: We consider the impact of AI-powered real-time traffic analysis and network resource optimization (Network Slicing) on the quality of autonomous driving services within the MEC architecture.

2.3. Final Target Outcomes: Deriving Technical Guidelines and Simulation Data

Through this report, our team aims to present the following specific deliverables.

End-to-end (E2E) latency analysis model: We will formulate the factors of latency on the integrated architecture where wired TSN and wireless 5G URLLC are combined and derive a latency budget for each segment.

Network performance verification data: We will use simulators such as NS-3 or OMNeT++ to generate data comparing performance (latency, jitter, packet loss) before and after applying LCN technologies in autonomous driving scenarios (e.g., platooning, remote driving).

Industry standard-based technical guidelines: We will propose technical recommendations for optimal network protocol configurations and fail-safe mechanisms by autonomous driving level.

3. Plan and Timeline

3.1. Research Methodology: Multi-layered Analysis and Simulation-Based Verification

This study adopts a multi-layered methodology combining theoretical analysis, practical case studies and technical verification.

Precise analysis of standards and white papers: We will understand technical operating principles based on the latest standard documents of IEEE, IETF and 3GPP. We will also collect best practices from real industrial settings through white papers from major equipment vendors such as Cisco, Ericsson and Huawei.

Component-based architecture design: We will design the entire network path connecting autonomous vehicles, base stations (gNB), edge servers (MEC) and core networks (5GC) by modules and analyze interoperability between modules.

Network simulation (NS-3): We will model the mobility and traffic patterns of autonomous vehicles and perform simulations similar to real environments. In particular, by integrating traffic simulator SUMO with NS-3, we will analyze the impact of communication environments on vehicle control.

3.2. Detailed Milestones and Implementation Plan (9 Weeks)

Week(s)	Stage	Main Activities	Target Deliverables
Weeks 1–2	Initial Research and Requirements Analysis	- Define network KPIs for each autonomous driving level - Analyze IEEE 802.1 TSN and 5G URLLC standards - Review relevant industry white papers and recent papers	Technical requirements definition document, literature analysis summary
Weeks 3–5	System Design and Simulation Setup	- Design an integrated LCN architecture (wired-wireless bridging) - Build NS-3 simulation environment and set parameters - Define traffic scenarios (e.g., platooning)	Network architecture diagram, draft simulation code
Weeks 6–8	Conduct Experiments and Prepare Final Report	- Run simulations and collect data - Analyze and visualize performance results (graphs/tables) - Draw conclusions and conduct internal peer review	Final interim report, simulation analysis report
Week 9	Final Review and Video Recording	- Conduct final verification of all	Final results demonstration

		simulations and review reliability of result data - Record a video demonstrating key project results and technical explanations	video, final term report

4. Individual Roles

The roles of the four team members are divided according to technical expertise and project management needs as follows.

Team Member Name (Student ID)	Main Role	Detailed Responsibilities and Duties
[전남규(2020271319)]	Wireless Protocol and URLLC Analysis	- Analyze 3GPP 5G/6G URLLC physical layer technologies - Analyze latency factors in the wireless segment and establish optimization strategies - Research 5G network slicing technologies and study application methods
[전예규(2025270607)]	Wired TSN and Standards Expert	- Conduct precise analysis of IEEE 802.1 TSN standards (including 802.1DG) - Design in-vehicle Ethernet architecture and study traffic control logic - Analyze wired/wireless synchronization (gPTP) interworking mechanisms
[조승찬(2022270620)]	L3 DetNet and Edge Architecture and Video Recording	- Research IETF DetNet (e.g., RFC 8655) architecture and Layer 3 deterministic paths - Design MEC-based latency optimization architecture and conduct case studies - Build cloud-interworking security and reliability framework - Record the final demonstration video

[조석호(2021270668)]	Simulation and Technical Documentation	- Perform NS-3-based autonomous driving network simulations and extract data - Prepare technical documentation and manage templates; monitor timeline and milestones - Manage project records (meeting minutes, etc.) and review the final report

However, the default assumption is that simulations will be performed collectively so that all members can do them.

5. Technical Deep Dive into Latency-Critical Networks

5.1. Paradigm Shift in In-Vehicle Network Architecture

Traditional in-vehicle networks have relied on low-speed communication methods such as CAN (Controller Area Network) or LIN (Local Interconnect Network). However, as autonomous driving levels rise and the amount of sensor data increases exponentially, in-vehicle networks are rapidly transitioning to high-speed Ethernet-based TSN. In particular, a zone-based architecture groups sensors and actuators according to their physical location in the vehicle and connects them via a high-speed Ethernet backbone, reducing wiring complexity and increasing data processing efficiency.

In this environment, **IEEE 802.1Qbv(Time-Aware Shaper)** acts as the 'traffic police' of the network. It divides the total communication time into periodic slots and allocates dedicated time windows (gate open) for critical control data, fundamentally eliminating collisions with other general data. Simulation studies show that such deterministic scheduling can keep the worst-case latency of mission-critical traffic within tens of microseconds.

5.2. Analysis of Integration Mechanisms of 5G URLLC and TSN

For autonomous vehicles to cooperate with external infrastructure, the in-vehicle TSN network must extend over the wireless 5G section. 3GPP implements this by treating the 5G system as a “virtual TSN bridge.” The most important elements in this integration process include the following.

Time synchronization: Autonomous vehicles and ground infrastructure must share the same time reference (grandmaster clock). The 5G system supports the IEEE 802.1AS protocol, synchronizing time in the wireless section with an error of less than one microsecond, which plays a key role in platooning where multiple vehicles cooperate to drive.

QoS mapping: Accurately mapping the Priority Code Point (PCP) of the TSN layer to the 5G layer's QoS flow (5QI) ensures that the wireless section provides the same level of priority handling as TSN.

5.3. Deterministic Networking across Wide Area via IETF DetNet

When an autonomous vehicle communicates with a remote control center or passes through wide-area networks across multiple cities, Layer 2 TSN alone has limitations. In this case, IP-based IETF DetNet technology is applied. DetNet reserves resources and fixes paths between IP routers, minimizing jitter even across multi-hop sections.

In particular, **Packet Replication and Elimination Function (PREOF)** dramatically improves reliability. The same control packet is sent along different paths (Path A and Path B), and the receiver adopts the packet that arrives first and discards the duplicate packet that arrives later. This technology enables uninterrupted data transmission even when the wireless section experiences temporary signal degradation or network equipment failures, ensuring the safety of autonomous driving.

Technical Standard	Key Mechanism	Expected Impact in Autonomous Driving
IEEE 802.1Qbv	Time-Aware Gate Control (TAS)	Ensures determinism of control command transmission time and eliminates collisions
IEEE 802.1Qbu	Frame Preemption and Interruption	Allows immediate insertion of important data during transmission of low-priority traffic
3GPP URLLC	Mini-slot and Grant-Free Access	Minimizes waiting time in the wireless segment (ms)
IETF DetNet	Layer 3 Resource Reservation and PREOF	Ensures reliability and low latency in wide-area and heterogeneous network segments
ETSI MEC	Forward Deployment of Computing Resources	Eliminates cloud processing delays and enables real-time sensor data analysis

5.4. Future Outlook: 6G and AI-Native Autonomous Networks

The upcoming 6G era is expected to bring even more intelligent autonomous driving networks. The network itself will function as an AI inference engine, analyzing sensing data from vehicles and guiding optimal driving routes to realize "network-assisted autonomous driving." In addition, integrated sensing and communication (ISAC) technology will allow base stations to detect surrounding vehicles and pedestrians directly like radar and send that information to vehicles through the network, overcoming the physical limitations of vehicle sensors (such as blind spots).

To manage and operate such sophisticated systems, **Intent-based Networking** is being introduced. When an operator inputs a high-level intent such as “maintain latency below 5 ms,” an AI agent dynamically adjusts the network’s slicing parameters and routing paths in real time to satisfy the condition. This will be a decisive technology for maintaining optimal network performance without human intervention in the complex and dynamic environment of autonomous driving.

6. Conclusion

This report has examined in detail the technical requirements that latency-critical networks in the autonomous driving sector must meet and the wired–wireless integrated architecture needed to implement them. To guarantee the safety and efficiency of autonomous driving, it confirmed that beyond merely enhancing the performance of individual technologies, the deterministic control of wired TSN, the ultra-fast transmission of wireless URLLC and the structural optimization of MEC and DetNet must be organically combined.

Through the proposed nine-week implementation plan, our team will design a network model that meets actual industrial standards and produce quantitative data through simulation, completing practical technical guidelines for building autonomous driving infrastructure. The technical insights and practical competencies gained in this process are expected to become a powerful asset demonstrating our expertise as next-generation network professionals.

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